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Topic 4: Natural Selection and Genetic Modification

Edexcel GCSE (9-1) Biology (1BI0) — Paper 2

4.1 Darwin's Theory of Evolution by Natural Selection

> **Evolution** is the gradual change in the inherited characteristics of a population over many generations, through the process of **natural selection**, which may result in the formation of new species.

In 1859, Charles Darwin published *On the Origin of Species*, presenting the theory of evolution by natural selection. Darwin spent five years sailing on HMS *Beagle*, observing living things across the globe — most famously in the Galápagos Islands — and built up a huge body of evidence. His co-worker **Alfred Russel Wallace** independently reached the same conclusions at the same time; their ideas were first announced together in 1858.

Darwin's theory explains how species change over time using four key observations:

The Four Steps of Natural Selection

1. **Variation** — individuals within a species show a range of inherited characteristics. This variation comes from random mutations in DNA, and is passed on through alleles.
2. **Competition (struggle for existence)** — more offspring are produced than the environment can support. Individuals compete for limited resources: food, water, mates, space and light.
3. **Survival of the fittest** — individuals whose characteristics are best suited to the environment are more likely to survive and reproduce.
4. **Inheritance** — these successful individuals pass their advantageous alleles to their offspring. Over many generations, the frequency of advantageous alleles in the population increases,

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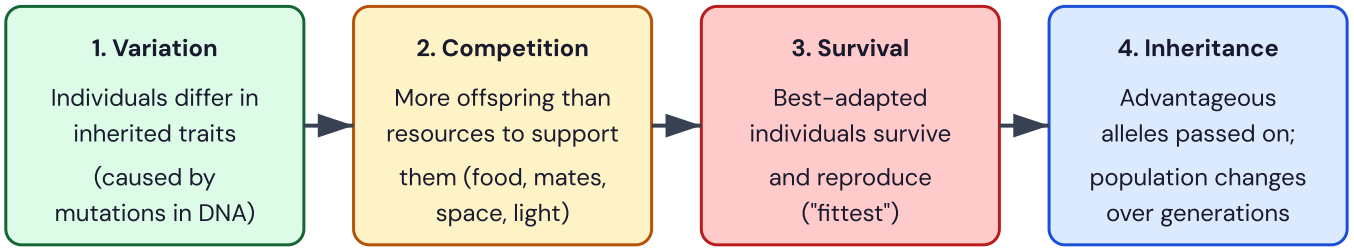
and the species changes.

Exam Tip: Natural selection acts on **phenotypes** (what you see), but only **inherited** characteristics can be passed on. A scar from an accident is not inherited, so it cannot evolve. A mutation in a gamete — a change to DNA — *can* be passed on.

The Four Steps of Natural Selection



Repeat over many generations → species changes → new species may form

Case Study: Galápagos finches. Darwin observed 13 species of finch across the Galápagos Islands, all descended from a single ancestor that arrived from South America. On each island, the finches faced different food sources. Finches born with beak shapes best suited to the local food (e.g. large crushing beaks for hard seeds, long slim beaks for insects in bark, short sharp beaks for cactus flowers) were more likely to survive and reproduce. Over thousands of generations, beak variation became fixed in each population, producing distinct species. Modern work by Peter and Rosemary Grant has shown measurable beak-size changes in the medium ground finch during droughts — evolution happening within a human lifetime.

Speciation

If two populations of the same species become **isolated** (e.g. separated by a mountain, river, or ocean) and experience different selection pressures, they gradually accumulate different alleles. Eventually the two populations are so different that they can no longer interbreed to produce fertile offspring — they are now separate **species**.

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4.2 Evidence for Evolution

Darwin's theory was initially controversial because it conflicted with the religious view that species were created separately and fixed. It was only accepted after a large body of evidence had accumulated. Today, evidence for evolution comes from four main sources:

Source of evidence	What it shows
Fossils	A timeline of how organisms have changed over millions of years — extinct species, transitional forms, and the gradual appearance of modern groups.
Antibiotic resistance in bacteria	Natural selection happening in real time (within years) — directly observable.
The peppered moth	Observed shift in moth colour during and after the Industrial Revolution — evolution within a human lifespan.
DNA / molecular evidence	Closely related species share more DNA than distantly related ones — supports common descent.

Fossil Progression

The fossil record shows a clear order: simple organisms appear first in the oldest rocks, with more complex forms appearing in younger rocks. Whales, for example, evolved from four-legged land mammals — a series of transitional fossils (*Pakicetus*, *Ambulocetus*, *Basilosaurus*) show legs gradually shortening and nostrils moving backwards to form a blowhole. Human evolution is similarly supported by fossils of *Australopithecus*, *Homo habilis*, *Homo erectus* and *Homo sapiens*.

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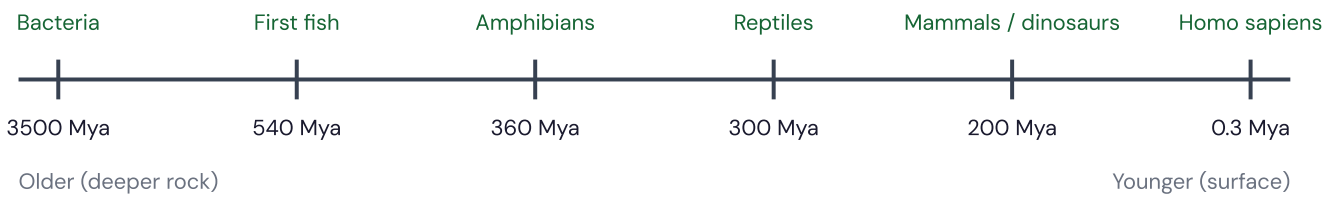
Simplified Fossil Record Timeline



Simple forms appear first; complex forms later — consistent with evolution over geological time

Case Study: The peppered moth (*Biston betularia*). Before the Industrial Revolution in Britain, almost all peppered moths were pale with small black speckles — camouflaged against light lichen-covered tree bark. A rare dark (melanic) form existed but was easily picked off by birds. During the 1800s, soot from factories killed lichen and blackened tree trunks. Now the pale form stood out and was eaten; the dark form was camouflaged. By 1895, 98% of peppered moths in Manchester were dark. Since the Clean Air Acts, trees have recovered and the pale form is once again common. The same species, the same gene — two rapid shifts in allele frequency driven by natural selection in response to environmental change.

4.3 Fossils

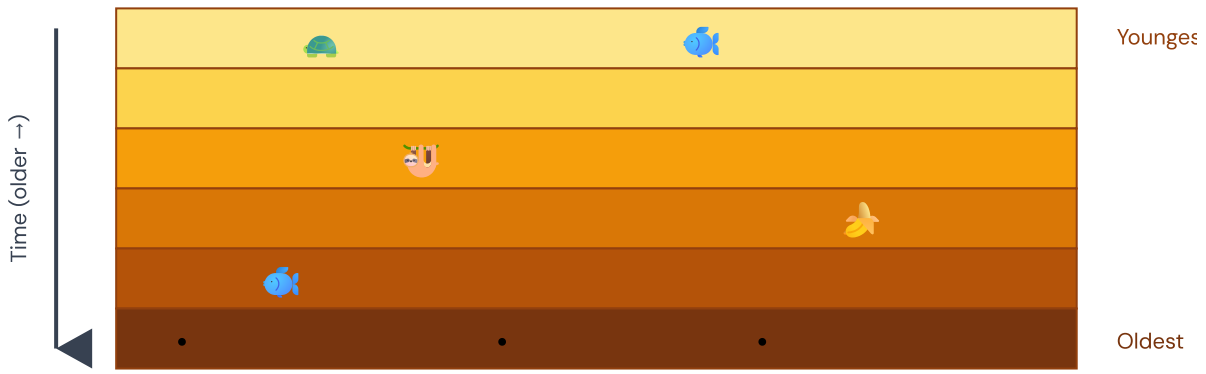
A **fossil** is the preserved remains, or evidence of, organisms from thousands or millions of years ago, usually found in sedimentary rock.

How Fossils Form

Method	How it works	Example
Hard parts preserved	Bones, teeth, shells do not decay easily. Buried in sediment, minerals gradually replace the original material, turning the hard parts to stone.	Dinosaur bones, ammonite shells

Casts and impressions	The organism rots away leaving a hollow (mould) in the rock. This may fill with minerals to make a cast. Footprints and burrows form impressions.	Trilobite moulds, dinosaur footprints
No decay (preservation)	Conditions are too cold, too dry, or lack oxygen, so decomposers cannot live. Whole bodies may be preserved.	Mammoths in permafrost; insects in amber; bog bodies in peat
Traces of activity	Footprints, burrows and droppings (coprolites) preserve evidence of behaviour even without the body.	Dinosaur trackways

Fossils in Sedimentary Rock Layers



Why the Fossil Record Is Incomplete

- **Soft-bodied organisms rarely fossilise.** Jellyfish, worms and most early life forms decayed completely before being buried, so we have few fossils of them.
- **Specific conditions are needed.** Most organisms die in environments where decomposers and oxygen destroy them before sediment can cover them.
- **Rocks are destroyed.** Geological activity (heat, pressure, erosion, subduction) destroys fossil-bearing rocks over millions of years.
- **Many fossils have not yet been found.** Most rocks are deep underground or under the sea.

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Exam Tip: A common exam point: "Why might the first forms of life not appear in the fossil record?" Answer: they were **soft-bodied** (no hard parts to fossilise) and the rocks from that era have mostly been **destroyed or deformed** over billions of years.

4.4 Antibiotic Resistance — Evolution in Action

Antibiotic resistance is the ability of a bacterium to survive exposure to an antibiotic that would normally kill it. Resistance arises by **mutation** and spreads by **natural selection**.

Antibiotics (such as penicillin, methicillin and streptomycin) kill bacteria or stop them reproducing, and have saved millions of lives since the 1940s. Resistance has evolved alarmingly quickly because bacteria reproduce very fast — a population can double every 20 minutes, so many generations pass in a single day.

How Resistance Evolves

1. A large population of bacteria contains random **mutations**. By chance, one bacterium has a mutation that makes it resistant to an antibiotic (e.g. it produces an enzyme that breaks down the drug, or changes the shape of the protein the drug targets).
2. When the antibiotic is used, the non-resistant bacteria are killed but the resistant one **survives**.
3. With its competitors gone, the resistant bacterium **reproduces rapidly** by binary fission, passing on the resistance allele to all its offspring.
4. Within days, the whole infection may be resistant — the antibiotic no longer works.

Case Study: MRSA (methicillin-resistant *Staphylococcus aureus*). *S. aureus* is a common skin bacterium that can cause serious wound and blood infections. In hospitals, the heavy use of the antibiotic methicillin killed off non-resistant strains — but strains with a mutated *mecA* gene (which produces a penicillin-binding protein the antibiotic cannot bind to) survived and multiplied. MRSA now causes tens of thousands of hospital deaths

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worldwide each year and must be treated with rarer last-line antibiotics such as vancomycin. Some strains have now developed resistance to vancomycin too (VRSA).

Slowing the Spread of Resistance

- Doctors avoid prescribing antibiotics for viral infections (antibiotics don't work on viruses anyway).
- Patients must complete the full course even if they feel better, so no partly-treated bacteria are left alive.
- Antibiotic use in agriculture (healthy farm animals) is being restricted.
- Pharmaceutical companies are investing in developing new antibiotics — but it takes 10+ years and is often not profitable.
- Hospitals enforce strict hand-washing and isolation procedures.

Exam Tip: Key language: antibiotics do **not** "cause" resistance. Mutations happen randomly, independently of the antibiotic. The antibiotic **selects for** already-resistant bacteria by killing the rest. Be careful never to write "the bacteria developed resistance because of the antibiotic" — this is the Lamarckian mistake.

4.5 Classification — The Linnaean System

Classification is the grouping of living organisms according to shared characteristics. The **Linnaean system**, devised by Carl Linnaeus in the 1700s, groups organisms into seven nested levels based on their physical features.

The Seven Levels

Level	Meaning	Human example
Kingdom	Broadest group (e.g. animals, plants)	Animalia

Phylum	Major body plans (e.g. backbone present)	Chordata
Class	Major groupings within a phylum	Mammalia
Order	Related families	Primates
Family	Closely related genera	Hominidae
Genus	Very closely related species	*Homo*
Species	Individuals that can interbreed to produce fertile offspring	*sapiens*

Exam Tip: Mnemonic: **K**ing **P**hilip **C**ame **O**ver **F**or **G**ood **S**oup — Kingdom, Phylum, Class, Order, Family, Genus, Species.

Binomial Naming

Linnaeus also introduced the **binomial system** of naming species: every organism has a two-part Latin name consisting of its **Genus** (capitalised) and **species** (lowercase), both in italics. For example:

- *Homo sapiens* — modern humans
- *Panthera leo* — lion
- *Panthera tigris* — tiger (same genus as lion, so closely related)
- *Canis lupus* — grey wolf
- *Canis lupus familiaris* — domestic dog (a subspecies of wolf)

Binomial naming solves the problem of common names: an "elk" in Europe is a different species from an "elk" in North America. The Latin name is universal and unambiguous, allowing scientists worldwide to communicate precisely.

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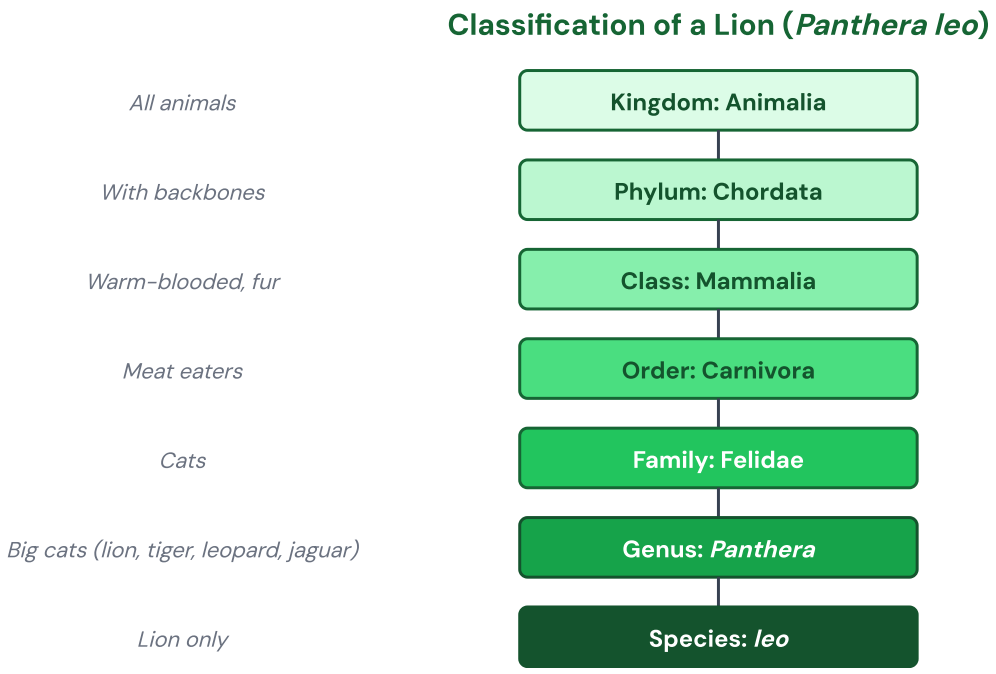

4.6 The Three-Domain System

For centuries, scientists used the Linnaean system with five Kingdoms: Animalia, Plantae, Fungi, Protista and Prokaryota (bacteria). In 1977, American microbiologist **Carl Woese** used a new technique — comparing the sequence of **ribosomal RNA (rRNA)** between organisms — and made a startling discovery. The "bacteria" were actually *two completely different groups* whose rRNA sequences were as different from each other as either was from a human. He proposed that all life should be divided into three **domains**:

Domain	Examples	Key features
Bacteria	*E. coli*, *Streptococcus*, cyanobacteria	True bacteria; prokaryotes; live everywhere — soil, water, body
Archaea	Methanogens, *Sulfolobus*, halobacteria	Look like bacteria but different biochemistry; often found in extreme environments (hot springs, salt lakes, deep-sea vents)
Eukarya	Animals, plants, fungi, protists	Cells have a true nucleus and membrane-bound organelles

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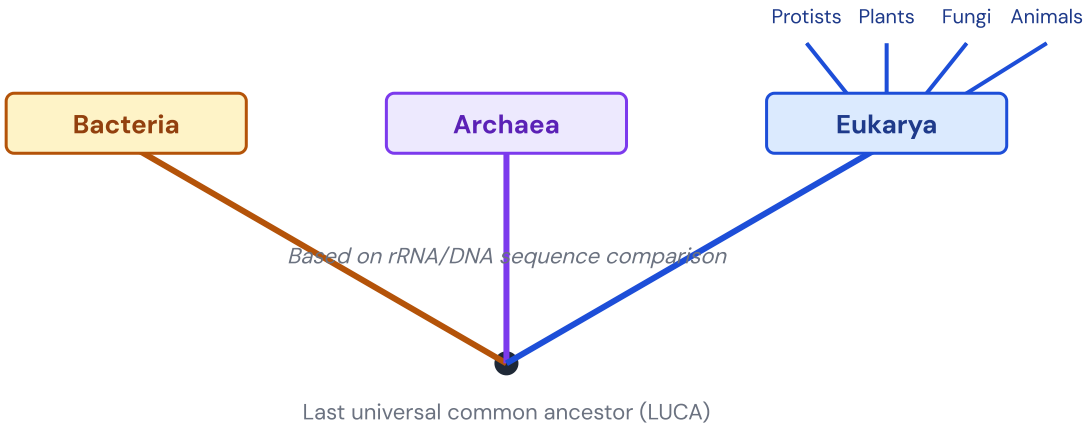
The three-domain system is now widely accepted because molecular evidence (DNA and rRNA) is much more powerful than physical appearance alone. It showed that some organisms we had grouped together were actually more distantly related than they appeared, while others were closer.

Woese's Three-Domain Tree of Life



Exam Tip: Remember the evidence for the three-domain system is **molecular** — differences in ribosomal RNA (rRNA) sequences. This was the first time classification was based on genetics, not just appearance.

4.7 Selective Breeding (Artificial Selection)

Selective breeding is the deliberate choice by humans of which organisms to breed together, so that offspring inherit desired characteristics. Over many generations, the population changes in a useful direction.

Humans have been doing this for at least 10,000 years, since we domesticated wolves into dogs and selectively bred wild grasses into wheat, barley and maize. The principle is identical to natural selection — except the "selection pressure" is the human farmer, not the environment.

The Process

1. Choose parents that best show the desired characteristic (e.g. highest milk yield, most disease-resistant crop).
2. Breed them together.
3. From their offspring, select those that most strongly show the desired trait.
4. Breed the selected offspring together.
5. Repeat over many generations — the trait becomes stronger and more uniform in the population.

Example	Desired trait selected for
Dairy cows (Holsteins)	High milk yield — modern dairy cows produce ~30 L milk per day, 10× more than 100 years ago
Wheat	Short stems (don't fall over), high yield, disease resistance (e.g. resistance to rust fungi)
Domestic dogs	Docile temperament, specific appearances, working abilities (herding, hunting, retrieving)
Racehorses	Speed, stamina, muscle development
Chickens	Fast growth (broilers), high egg laying (layers)
Tomatoes	Larger fruit, uniform ripening, longer shelf life

Problems with Selective Breeding

The advantage of selective breeding — concentrating useful alleles — is also its main problem: it **reduces the gene pool** (the range of alleles in the population).

- **Reduced genetic variation** makes the species vulnerable. If a new disease appears, there are few resistant individuals. An entire crop or herd can be wiped out.
- **Inbreeding depression.** Breeding close relatives together (common to keep a trait "pure") brings together harmful recessive alleles, leading to health problems.
- **Pedigree dog health issues.** Bulldogs bred for flat faces suffer breathing difficulties; German Shepherds bred for sloping backs develop hip dysplasia; Cavalier King Charles Spaniels have

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skulls too small for their brains (syringomyelia); Dalmatians commonly develop deafness and bladder stones.

- **Animal welfare concerns** — many "breed standards" prioritise appearance over the animal's quality of life.
- **Loss of useful wild alleles** that might help adapt to future environmental changes.

4.8 Genetic Engineering

> **Genetic engineering** (also called genetic modification or GM) is the direct transfer of a specific gene from one organism into the DNA of another. The recipient organism — called **transgenic** — then expresses the new gene and produces the associated protein.

The Key Steps

1. **Identify and isolate the gene** of interest from the donor organism (e.g. the human gene coding for insulin).
2. **Cut out the gene** using a **restriction enzyme**, which recognises a specific DNA base sequence and cuts at that point. It leaves "sticky ends" — short, unpaired DNA sequences that can bond to complementary sticky ends.
3. **Prepare a vector.** A vector is a piece of DNA used to carry the new gene into the host cell. The most common vector is a **plasmid** — a small circular piece of bacterial DNA. The plasmid is cut open using the *same* restriction enzyme, so it has matching sticky ends.
4. **Insert the gene into the vector.** The sticky ends of the gene and the plasmid pair up; an enzyme called **DNA ligase** joins the sugar-phosphate backbones, sealing the gene into the plasmid. This is now **recombinant DNA**.
5. **Insert the vector into the host** (e.g. a bacterium). The bacterium takes up the plasmid through its cell wall.
6. **Select and grow** the modified bacteria, which now produce the new protein (e.g. insulin).

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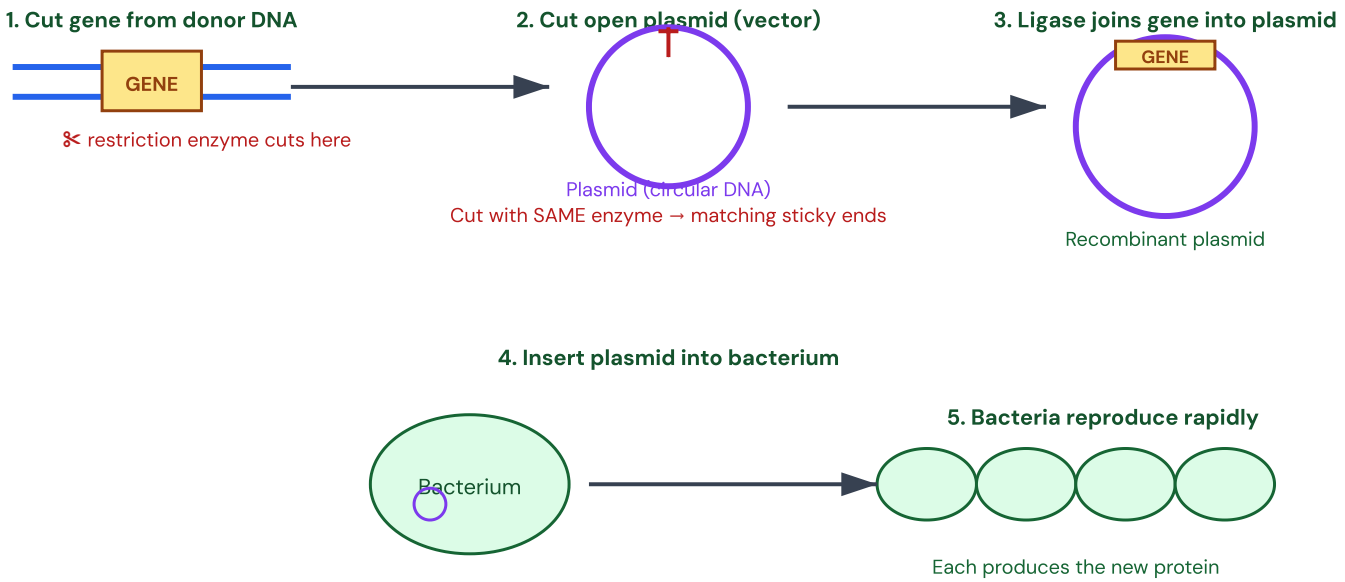
Genetic Engineering: Inserting a Gene into a Plasmid



Why Use the Same Restriction Enzyme?

Restriction enzymes cut DNA at *specific base sequences*. Each enzyme leaves a characteristic "sticky end" — a short single-stranded overhang of bases. For the gene to fit into the plasmid, the sticky ends must be complementary. Using the same enzyme on both ensures this. Ligase then forms the covalent bonds to seal the join permanently.

4.9 Examples of Genetic Modification

Organism	Gene inserted	Benefit
GM bacteria (*E. coli*) producing human insulin	The human insulin gene	Pure human insulin for diabetics; cheap, ethical (before GM, insulin was extracted from pig pancreases)
Golden Rice	Genes from daffodils/maize + a soil bacterium that cause the rice	Prevents childhood blindness from vitamin A deficiency in

	to make beta-carotene (vitamin A precursor)	countries where rice is the staple food
Bt maize / Bt cotton	Gene from *Bacillus thuringiensis* that produces an insect-killing protein (Bt toxin)	Plants make their own insecticide — fewer pesticide sprays, higher yield
Herbicide-resistant crops (Roundup-Ready soya, maize)	A bacterial gene giving resistance to the herbicide glyphosate	Farmers can spray herbicide to kill weeds without harming the crop
GM salmon (AquAdvantage)	Growth hormone gene from a Chinook salmon under the promoter of another fish gene	Grows to market size in half the time
GM tomatoes (Flavr Savr)	Antisense gene silencing a softening enzyme	Longer shelf life

Case Study: Insulin from GM bacteria. Before 1982, people with type 1 diabetes relied on insulin extracted from the pancreases of slaughtered pigs and cows. It was expensive, slightly different from human insulin, and caused allergic reactions in some patients. Scientists isolated the human insulin gene, inserted it into a plasmid vector using restriction enzymes and ligase, and transferred the plasmid into *E. coli*. The bacteria grow in huge fermenters and produce pure human insulin, which is filtered out and purified. This was the first genetically engineered medicine approved for human use and transformed diabetes treatment.

4.10 Benefits and Concerns of Genetic Modification

Benefits

- **Higher yields** feed more people. GM crops can be pest- or drought-resistant.
- **Improved nutrition.** Golden rice supplies vitamin A, reducing childhood blindness.

- **Reduced pesticide use.** Bt crops produce their own insecticide — less spraying, lower cost, less chemical pollution.
- **Medical benefits.** Pure insulin, clotting factors, growth hormone and vaccines from GM bacteria.
- **Longer shelf life** — less food waste.
- **Ability to grow in poor conditions** — salt-tolerant, drought-tolerant varieties could farm previously unusable land.

Concerns

- **Gene transfer to wild species.** Pollen from herbicide-resistant crops could cross-pollinate with wild relatives, producing "superweeds".
- **Harm to non-target species.** Bt toxin might affect beneficial insects (e.g. monarch butterflies feeding on milkweed near Bt maize fields).
- **Reduced biodiversity.** Farms growing one GM variety leave no room for wild plants or the insects that feed on them.
- **Allergens.** A gene transferred from a nut to another crop could carry the allergen into a food people thought was safe.
- **Long-term human health effects** are still debated; most evidence suggests approved GM foods are safe.
- **Ethics.** Concerns about "playing God", animal welfare (GM animals), and large corporations controlling the global seed supply.
- **Farmer dependency.** Seeds from GM crops are often patented; farmers must buy new seeds each year rather than saving their own.

> **Exam Tip:** Exam questions often ask you to "evaluate" GM. A top-mark answer must include specific benefits, specific concerns, and a reasoned conclusion — not just opinion. Example conclusion: "GM crops can significantly improve yield and nutrition, and their benefits outweigh the risks *provided* they are carefully tested and regulated."

4.11 Tissue Culture in Plants

Tissue culture is the growth of new tissues or whole organisms from small samples of cells in a sterile nutrient medium. In plants, this produces genetically identical **clones** of the parent.

The Method (plants)

1. Take a small piece of tissue — the **explant** — from a parent plant (usually from a fast-growing region such as a meristem at a shoot tip).
2. Sterilise the explant with dilute bleach to kill any bacteria and fungi.
3. Place the explant on a sterile **agar growth medium** containing nutrients (sugars, minerals) and **plant growth hormones** (auxin, cytokinin).
4. The cells divide by mitosis and form a mass of undifferentiated cells called a **callus**.
5. Changing the ratio of hormones triggers the callus to develop roots and shoots, producing tiny plantlets.
6. The plantlets are transferred to soil and grow into full plants, all genetically identical to the parent.

Why Use Tissue Culture?

- **Conservation of endangered plants.** Orchids and other rare plants can be multiplied from a single specimen without damaging wild populations.
- **Commercial horticulture.** Growers produce thousands of identical plants (e.g. strawberry, banana, orchid) with the desired traits.
- **Disease-free stock.** Starting from a meristem produces virus-free plants.
- **Fast propagation.** Much faster than growing from seed.
- **Year-round production** in sterile labs, independent of season.

Exam Tip: Key word: **clone**. Tissue culture produces genetically identical copies, because it uses **mitosis**. This means the offspring inherit every useful trait of the parent — but there is no genetic variation, and a single disease could wipe out the entire crop.

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4.12 Tissue Culture in Animals

Animal cells can also be grown in culture, though the process is more difficult than with plants because animal cells lose the ability to differentiate much earlier than plant cells do.

Uses of Animal Tissue Culture

- **Medical research.** Growing human cells in dishes allows scientists to test drugs, study cancer, and investigate how cells respond to infection without using live animals.
- **Producing monoclonal antibodies** — identical antibodies used in pregnancy tests, cancer treatment, and diagnostics.
- **Stem cell therapy** — the potential to grow replacement tissues (skin, cartilage, retinal cells, beta cells for diabetes) from stem cells.
- **Studying genetic disorders** — cells from affected individuals are grown and compared with healthy cells.
- **Reducing animal experiments** — cell cultures can replace some uses of live animals in toxicology testing.

Comparison	Plant tissue culture	Animal tissue culture
Main purpose	Commercial cloning; conservation	Research; medical therapy
Outcome	Whole new plants	Sheets of cells; sometimes engineered tissues
Ease	Relatively easy — plant cells remain totipotent	Harder — most animal cells cannot redifferentiate
Hormones needed	Auxin, cytokinin	Specific growth factors

4.13 Fertilisers and Biological Control

To produce enough food for a growing world population, farmers use a range of techniques to boost yield. Two of the most important are fertilisers and pest control.

Fertilisers

Plants need mineral ions from the soil to make essential biological molecules. When crops are harvested, these ions are removed with the plant and not returned to the soil, so they must be replaced.

Mineral	Required for	Deficiency symptom
Nitrogen (as nitrate ions NO_3^-)	Making amino acids → proteins (for growth, DNA, enzymes)	Poor growth, yellow leaves (chlorosis)
Phosphorus (as phosphate ions PO_4^{3-})	Making DNA and cell membranes; root growth	Poor root growth, discoloured (purplish) leaves
Potassium (as K^+ ions)	Enzyme function; flowering and fruit production	Yellow leaves with dead patches; poor fruit

Fertilisers may be:

- **Artificial (inorganic)** — manufactured, releases ions fast, precise N:P:K ratios (e.g. "NPK 15:10:20"). Can leach into rivers and cause eutrophication.
- **Natural (organic)** — manure, compost, bone meal. Releases nutrients slowly as it decomposes, also improves soil structure.

Biological Control vs Chemical Pesticides

Biological control uses a living organism (a natural predator, parasite or disease) to reduce the population of a pest. **Chemical pesticides** are synthetic chemicals sprayed onto crops to kill pests directly.

	Chemical pesticides	Biological control

Example	Organophosphates sprayed on aphids	Releasing ladybirds to eat aphids; parasitic wasps on whitefly
Speed of effect	Fast — kills pests within hours	Slow — takes weeks as predator population grows
Specificity	Often kills beneficial species too (e.g. bees)	Usually species-specific
Persistence	Chemicals may accumulate in food chains (bioaccumulation)	Self-sustaining; no residues in food
Resistance	Pests evolve resistance; pesticide stops working	Unlikely — predator evolves alongside
Risks	Toxicity to humans, pollinators and wildlife	Introduced species may become pests themselves (e.g. cane toads in Australia)
Cost	Must be reapplied each season	One-off — cheaper long-term

Case Study: Cane toads in Australia. In 1935, cane toads (*Rhinella marina*) were introduced to Queensland as a biological control for beetles damaging sugar cane crops. The toads barely touched the beetles — but thrived on native wildlife. They now number hundreds of millions, have spread across northern Australia, and their toxic skin has killed countless native predators (quolls, goannas, snakes) that try to eat them. A warning that biological control can backfire if the introduced organism has no natural predators.

Integrated Pest Management

Modern farmers often use a combination: crop rotation, resistant varieties, some biological control, and targeted pesticide use only when needed. This approach balances high yields with environmental protection.

Summary of Topic 4: Darwin's theory of natural selection explains how variation + competition + survival + inheritance change species over generations. The evidence is overwhelming: fossils, antibiotic resistance, the peppered moth and molecular data. Classification organises life using the Linnaean system (K-P-C-O-F-G-S) with binomial names; Woese's three-domain system (Bacteria, Archaea, Eukarya) uses molecular data. Humans also shape species by selective breeding (slow, traditional) and genetic engineering (fast, precise — insulin, golden rice, Bt maize). Tissue culture clones plants and grows animal cells for research and therapy. Fertilisers (N, P, K) and biological control are used to maximise food production — but each has costs as well as benefits.

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